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Ingestion Team

AI-Led Development

Environment

Claude Code + GitHub Copilot + New Relic MCP

🚀 10× faster feature delivery

🧠 Onboard in hours, not weeks

🔍 30-min incident diagnosis

⚡ 5 agents working in parallel

📦 Knowledge that compounds

The Challenge — Four Real Pains the DataOS Ingestion Team Was Living With

🕐 New Engineer Onboarding

2+ WEEKS

to reach full productivity in the codebase

🔍 Incident Root Cause

1–10 HOURS

backtracking pipelines one by one manually

💥 Hidden Blast Radius

5-DAY

incident from what looked like a '1-day fix'

🗂️ Context After Absence

HOURS–DAYS

to rebuild codebase understanding after a break

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DataOS Ingestion — How We Run AI-Led Development Today

What We Own

om-airflow-dags

OpenMetadata ingestion pipelines
(Airflow DAGs, connectors, plugins)

spark-kafka-apps

Spark + Kafka streaming pipelines
(real-time ingestion to warehouse)

tf-dataos-new-relic

Terraform NR monitoring
(alerts, dashboards — 55 commits/month)

Daily Workflow

1. **Open Claude Code in repo**
   Skills auto-load context + graph
2. **GitHub Copilot in IDE**
   Line completion as we code
3. **Work — skills auto-trigger**
   /airflow /pyspark /nrql /terraform...
4. **NR MCP for incidents**
   Query logs, metrics, traces in-chat
5. **/avengers for complex tasks**
   Parallel agents — 1 day → 2 hours
6. **/wrap-up at session end**
   Jira updated · lessons captured · graph refreshed

Tools in Use

Claude Code
Team environment
+ 20 skills

GitHub Copilot
Line-level AI
in IDE

NR MCP
Live observability
12 tools

Jira + Confluence
Auto-updated
every session

Code Graph
Blast radius +
impact analysis

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Two AI Tools. Different Jobs. Better Together.

GitHub Copilot — Line-level AI

- ✅ Autocomplete lines & functions in IDE
- ✅ Inline chat: explain or refactor code
- ✅ Generate tests & boilerplate fast

- — No cross-session codebase memory
- — Can't query NR, Jira, Confluence
- — No impact / blast-radius analysis
- — No multi-agent parallel work

+

Together: 8–12×

Claude Code — Team environment AI

- ✅ Full codebase context at every session start
- ✅ Queries NR logs, metrics, alerts in real time
- ✅ Knows what breaks before you write a line
- ✅ 5 parallel agents on one task with /avengers
- ✅ Replayable golden patterns from past fixes
- ✅ 20 skills auto-trigger on file type or topic
- ✅ Jira + Confluence auto-update each session
- ✅ 5-layer code review: CC + Copilot + agents + human

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System Architecture — How It All Connects Under One Environment

Claude Code AI-Led Dev Environment — v2

hot/cold memory · code-review-graph · golden/replay · budget dial

Session Boot #cce4f7

→ Claude Code starts

loads

~/.claude/CLAUDE.md
global rules + routing + skills

loads

<repo>/.claude/CLAUDE.md
only 2 @ refs
@ hot.md + @ GRAPH_REPORT.md (~4k tokens total)

pulls on demand

deeper memory via MCP (not @-loaded)

~4k tokens at boot (3-7× cheaper than v1)

fires on every command

Hooks Layer #fdf3c0

rtk hook claude
Sub: PreToolUse → every Bash call
Badge: 60-90% token savings

refresh-mcp-tokens.py
Sub: UserPromptSubmit → MCP OAuth

log-grep-usage.sh NEW
Sub: PostToolUse Grep|Glob → telemetry
→ weekly review tunes graph coverage

self-heal-stop.sh
Sub: Stop → lesson-capture reminder

stop-record-cost.sh NEW
Sub: Stop → append session $ to cost.jsonl
→ feeds /budget status

statusline.sh
Sub: model · context % · session $ · daily

IN hot.md auto-loaded only

/bootstrap seeds on first visit

/wrap-up regenerates hot.md each session end

self-heal-stop auto-captures corrections

memory MCP deeper fetches pull-on-demand

Per-Project Memory (hot/cold)

~/.claude/projects/<repo>/memory/

AUTO-LOADED on boot
[hot.md ~2k] Big Bold
curated digest, wrap-up regenerates

pull on demand via MCP
[MEMORY.md] [architecture.md]
[todo.md] [lessons.md] [history.md]
mistakes → append → 3× = Pattern rule
self-improvement loop

~/.claude/projects/<repo>/plans/
Sub: mirrored from ~/.claude/plans/ at /wrap-up

/golden + /replay NEW ZONE #d4c4f0

the compounding layer

/golden save <slug>
distills validated session
writes ~/.claude/golden/<slug>.md
Schema
pattern | scope | tags | trigger_hints
Symptom → Root Cause → Steps That Worked
→ What NOT To Do → Files Touched

/replay <slug>
validates staleness
(<30d, files exist)
loads golden prior-art
50-80% cheaper than re-deriving

~/.claude/golden/
index.json
N patterns · tagged · dated

IN /wrap-up Phase 3.5 auto-prompts on save-worthy sessions [y/n/edit]

reads CLAUDE.md + repo context

reads todo.md + lessons.md context

OUT mission complete → trigger wrap-up

First Visit Only — /bootstrap

(run once per new repo, Opus + high effort)

Phase 1 → **detect-stack.sh**
→ framework, framework, primary}

Phase 2 → **seed-memory.sh**
→ 6 files + plans/ dir

Phase 3 → **write-project-claude.sh**
→ .claude/CLAUDE.md + .mcp.json + .gitignore

Phase 4 → **build-graph.sh** NEW
replaces /graphify
- code-review-graph install (wire MCP)
- code-review-graph build (SQLite graph)
- compose GRAPH_REPORT.md (from wiki + status)

Phase 5 → **populate architecture.md**
← from GRAPH_REPORT.md

OUT seeds hot.md + 5 cold files + plans/

OUT builds SQLite graph

OUT wires both servers in .mcp.json

OUT @ hot.md + @ GRAPH_REPORT.md

/avengers — Multi-Agent Orchestration #daeeff

Fury (Opus) orchestrates specialists (Sonnet) in parallel

Nick Fury — Captain
Badge: model: opus
reads mission from Jira/context
writes plan → avengers-{TS}-plan.md
TeamCreate + TaskCreate upfront
Arms ScheduleWakeup(90s) monitoring
Routes C/R/T/V
Respawns crashed agents autonomously
crashed agents → respawn

Coders — 3 always
[stark-senior]
[stark-techlead]
[stark-seasoned]
Badge: model: sonnet
Label: parallel, different task batches

Pipeline Specialists
[natasha-reviewer]
→ [banner-tester] →
[hawkeye-validator]
Labels: code review | run tests | final gate
model: sonnet

State File
/tmp/avengers-{TEAM}.json
phase · agents · tasks · blocked

Dashboard
bridge.py :2026
https://avengers:2026 · live mission view

Optional Specialists (tiny)
strange-architect | rogers-data-engineer | maximoff-python | thor-devops
spawned on-demand by Fury

Processing Pipeline #d4edda

TOP #fdc060
code-review-graph, wa graphify
Sub: Tree-sitter AST + SQLite + 20 f9 tools
Location: <repo>/.code-review-graph/-(gitignored)
28-tools semantic_search_nodes · query_graph
get_impact_radius · detect_changes · get_review_context ·
get_architecture_overview · get_hub_nodes · get_bridge_nodes...
auto-updates on every Edit via hook (<2s)

MIDDLE #c4a8f0
memory MCP server NEW
Sub: pull-on-demand access to cold memory
Location: ~/.claude-kit/mcp-servers/memory-serr/
5 tools get_memory · search_memory · get_lessons
get_todo · recall_plan
no more 20k @-loads

BOTTOM
/wrap-up
Sub: session end ritual — 6 phases
0: corrections audit
1: append-history
2: todo update
3: lessons dedupe + CLAUDE.md emotion
3.5: golden-worthiness auto-prompt NEW
4: refresh-graph-report (OMG auto-update)
5: mirror-plans + regenerate plans NEW
6: handoff summary

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Full AI-Led Development Lifecycle — Time Savings at Every Phase

	01 Requirement	02 Spec & Plan	03 Architect	04 Build	05 Review	06 Test & Gate	07 Deploy	08 Monitor	09 Document	10 Learn
	Manager / PM	AI-assisted	AI + Graph	Task & volume dep.	5-layer AI+Human	AI auto-mated	Human initiates	NR MCP (AI)	AI auto-update	Golden patterns
Before:	—	4–6h	2–4d	2–3d	—	—	—	1–10h	manual	—
After:	—	20m–1h	1–2d	2–6h	—	—	—	0–20m	auto	auto

Confirmed by engineer · Spec+Arch+Build: days → hours · Kafka diagnosis: hours → 0–20 min · Overall: 8–12×

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Onboard in Hours, Not Weeks — Full Context on Day One

BEFORE

2 WEEKS

50 Slack DMs · git archaeology · tribal knowledge
Still not fully productive by end of week 2

→

AFTER

2 HOURS

/bootstrap: detects stack, seeds memory, builds graph
Full codebase context ready in 15 seconds

Full context Day 1

Architecture + lessons + todos
auto-loaded every session

Never re-explain the codebase

Team knowledge lives in
memory, not in one person

4 weeks/year saved

2 new engineers × 2 weeks
= 4 engineering-weeks back

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/avengers — Five AI Engineers Working in Parallel on One Task

Tasks that take one engineer one full day, completed in under 2 hours:

Nick Fury (Opus — Orchestrator)

routes work · validates gates · blocks bad code · shuts down on completion

Planner	Coder ×N	Reviewer	Validator
Researches codebase → implementation plan	Parallel code writing verifies build passes	Code standards + OWASP security check	Tests + impact analysis before any merge

1. Fury reads spec → spawns Planner + Coders simultaneously
2. Each agent works on an independent file — zero waiting
3. Reviewer + Validator run in parallel on all output
4. Human sees a single clean PR to approve

1 DAY

↓

2 HOURS

Airflow DAG + dbt model + NR alert — all in parallel

Engineer reviews 1 PR instead of 5 files

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Incident Investigation — AI Queries New Relic Directly, Instantly

BEFORE — Manual hunt

2–3 HOURS

Log in to NR UI · navigate to entity · scroll logs
switch tabs · compare charts · Slack teammates

Form hypothesis → test → repeat → still uncertain
Customer impact growing every minute

→

AFTER — AI-powered diagnosis

30 MINS

"analyze kafka consumer lag last 4 hours"
Claude queries NR, returns structured analysis

Consumer lag · partition skew · throughput drop
Affected services identified — fix scoped immediately

12 New Relic tools · Jira auto-updated · root cause in minutes

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Impact Analysis — Know What Breaks Before Writing a Single Line

We see the full impact of every change BEFORE it touches production.

⚠ Before: Invisible blast radius

Engineer changes spark_ingest.py.
Deploys. Downstream Kafka consumer breaks silently.
5-day incident. ~$50k in eng-hours to resolve.

→

✅ After: Instant clarity

Impact analysis runs before coding.
Sees 3 callers, 1 downstream consumer.
Fix scoped. Ships confidently. Zero incident.

~1 sprint/qtr

absorbed in surprise rework — before

< 1 day

rework after impact-aware development

0 blind deploys

every change impact-checked first

Auto-updates

dependency map rebuilt on every file save

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Documentation That Writes Itself — Jira & Confluence Always Up to Date

Work session completes → /wrap-up fires (~30s) → Claude drafts summary → Jira ticket auto-updated → Confluence published

Jira — What gets auto-updated

Status	In Review → Done · automatically
Comment	What was done + decisions made
Worklog	Time logged from session duration
Links	Commits linked via branch name

Confluence — What gets published

Architecture	Decisions logged after every session
Postmortem	Incident page auto-created on wrap-up
Release	Notes from git history — auto-versioned
Runbooks	Updated as team learns new patterns

Zero manual effort. Every session documented. Every time.

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Knowledge That Compounds

Every session makes every future session smarter — knowledge never leaves the team

ALWAYS LOADED

hot.md — ~2k tokens, auto-loads every session start

Active priorities + recent lessons + architecture summary. /wrap-up regenerates it. Zero manual effort.

PULL ON DEMAND

lessons.md · architecture.md · history.md · plans/

Team knowledge retrieved only when relevant — not loaded upfront. Cheaper, faster, no stale context drowning.

REPLAYABLE WINS

/golden save → /replay → 50× faster on repeat problems

After each validated fix: distill it into a golden. Next time the problem recurs: /replay loads the proven steps.

Month 1: 8 issues per task → Month 3: 1–2 issues. The team gets sharper, not staler.

The Results — Numbers the Team Has Measured

8–12×

overall productivity
for DE tasks

2–6h

dev ticket
vs 2–3 days before

0–20m

Kafka/NR diagnosis
vs 1–10 hours before

25+ SP

sprint capacity
+ adhoc on top

More measurements

5–15 min

context reset after absence
(vs hours/days before)

3–5h

new pipeline (full cycle)
vs 2–4 days before

4–6h → 20m–1h

spec & plan phase

2–4d → 1–2d

architecture + CDD
+ Confluence

8–12× overall for DE tasks · All metrics reported directly by Ingestion Team engineers

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Before & After — Ingestion Team · 3 Repos · 3 Months

What We Measured	Before	After (AI-Led)
Spec & Plan	4–6 hours	**20 min – 1 hour** ✅
Architecture + CDD	2–4 days	**1–2 days** ✅
Development ticket	2–3 days	**2–6 hours (task-dependent)** ✅
New pipeline (full cycle)	2–4 days	**3–5 hours** ✅
Kafka / NR diagnosis	1–10 hours	**0–20 minutes (NR MCP)** ✅
Context after absence	Hours / days	**5–15 min via /bootstrap** ✅
Sprint throughput	25 SP (at capacity)	**25 SP + additional + adhoc** ✅
Documentation effort	Manual, often skipped	**Automatic via /wrap-up** ✅
Code review layers	Manual review only	**CC Opus + Copilot + Natasha + Human** ✅

✅ = confirmed by Ingestion Team engineers · Repos: om-airflow-dags · spark-kafka-apps · tf-dataos-new-relic

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AI Does the Work. Humans Stay in Control. Always.

1	**Requirements**	Human writes the spec — AI reads it, asks clarifying questions
2	**Architecture**	Human approves the implementation plan — before a line is written
3	**Security Review**	AI flags every OWASP finding — any critical issue blocks the pipeline
4	**PR Merge**	Human reviews every PR — agent output treated like any other code
5	**Deploy**	Human initiates the deploy — AI writes release notes + monitors

Agent PRs treated identically to human PRs — same review, same standards, same gates.

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Getting Started — From Zero to Productive in 4 Steps

15 MIN

Step 1 — Install

One-time per engineer

install.sh checks prereqs, installs 20 skills, hooks, MCPs, budget dial — fully automated.

15 SEC

Step 2 — /bootstrap

First visit to any repo

Detects stack, seeds memory, builds code graph,
wires all MCPs — runs once, lasts forever.

ALWAYS

Step 3 — Work

Skills auto-trigger

20 skills activate on file type or topic. Copilot handles lines. Claude handles investigations.

30 SEC

Step 4 — /wrap-up

Every session end

Persists history, updates Jira, refreshes graph,
rebuilds hot.md. Habit that compounds daily.

Week-by-week rollout:

Week 1

Install + /bootstrap 1 repo + /wrap-up daily

Week 2

All repos + first /golden save after a win

Week 3

/avengers on a real multi-file task

Week 4+

Team golden library grows · 15–25× steady

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What We're Asking

Support adoption	Drive AI-led workflows across all Data Engineering & Management in DTM.
Allocate time	Engineers need 30 min/day: /wrap-up + goldens. The compound returns are real.
Endorse the model	Establish this as the standard template for all DE teams at New Relic.

15–25×
sustained avg

10×
novel work

50×
repeat tasks

30 min
to root cause

20–40×
ROI

Thank You · Questions?

Repo: github.com/nr-mdakilahmed/claude_code_setup